

Pi Plant Watering System(PPWS) Project Citation

Website: <https://plant-project-doc.vercel.app>

Github Link for Documentation: <https://github.com/nullzxc/Plant-Project-Documentation>

Description

This project implements an automated closed-loop control system for plant maintenance using a Raspberry Pi. Written in Python, the application utilizes multi-threading to handle concurrent tasks, including reading capacitive soil moisture sensors, controlling a 5V relay for water distribution, and capturing scheduled images via the Pi Camera. The software includes a user-friendly GUI developed with Tkinter and integrates IMAP/SMTP protocols to enable remote monitoring and command execution via email.

References & Citations:

Raspberry Pi Plant Watering

https://youtu.be/_NTW0npN4N0?si=fDICrYVT0FoDWjBK

Soil Sensor & Pump Material

<https://a.co/d/7AgVEt8>

Tkinter GUI

<https://www.geeksforgeeks.org/python/tkinter-colors/>

<https://www.w3resource.com/python-exercises/tkinter/python-tkinter-file-operations-and-integration-exercise-6.php>

<https://www.geeksforgeeks.org/python/python-creating-a-button-in-tkinter/>

Python Libraries

<https://github.com/mjs/imapclient>

<https://datasheets.raspberrypi.com/camera/picamera2-manual.pdf>